The log S-fBM Model: a Stationary Bridge between Rough and Multifractal Volatility

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Abstract

Empirical volatility is *rough*: its logarithm behaves like a fractional Brownian motion with Hurst exponent $H \approx 0.1$ or smaller. Multifractal random measures (MRM) describe the limit $H \to 0$ but have no roughness parameter. The log stationary fractional Brownian motion (log S-fBM) contains both. We state its construction, prove that its covariance is admissible, show that it converges to the log-normal MRM as $H \to 0$ at fixed intermittency, derive exact second-order moments of the integrated variance and of its logarithm, and explain why a generalised method of moments (GMM) built on these moments avoids the upward bias of scaling-based estimators of H.

1. Rough and multifractal volatility

Let S_t be a price with $d\ln S_t = \sigma_t\, dW_t$. Two empirical facts dominate high-frequency data: (i) $\mathbb{E}|\ln\sigma_{t+\tau} - \ln\sigma_t|^q \propto \tau^{qH}$ with $H \ll 1/2$ [2], which motivates rough fractional stochastic volatility (RFSV); (ii) the moments of the integrated variance scale non-linearly, $\mathbb{E}\big[(\int_t^{t+\tau}\sigma_s^2 ds)^q\big] \propto \tau^{\zeta(q)}$ with concave ζ, which motivates the MRM of Bacry, Delour and Muzy [1], a Gaussian multiplicative chaos [4] with logarithmic covariance. RFSV has no multifractality, MRM has no roughness parameter; log S-fBM [8] unifies them.

2. The S-fBM process

Definition 1. *Let $H \in (0, \frac12)$, $T > 0$, $\lambda > 0$ and $\nu^2 = \lambda^2/(H(1-2H))$. The S-fBM $\omega_{H,T}$ is the stationary Gaussian process with mean m and covariance*

$$\rho(\tau) = \tfrac{\nu^2}{2}\big(T^{2H} - |\tau|^{2H}\big)_+. \tag{1}$$

Lemma 1 (admissibility)**.** *ρ is positive definite on $\mathbb{R}$ for every $H \in (0, \frac12]$.*

Proof. ρ is even, continuous, non-increasing on $[0,\infty)$, tends to 0, and is convex on $(0,\infty)$: on $(0,T)$, $-|\tau|^{2H}$ is convex because $2H \le 1$, and at $\tau = T$ the derivative jumps from $-\nu^2 H T^{2H-1} < 0$ to 0, which preserves convexity. By Pólya's criterion [6], $\rho/\rho(0)$ is a characteristic function, hence positive definite (Bochner). □

The correlation vanishes beyond the *decorrelation scale* T, and for $|\tau| \ll T$, $\mathbb{E}(\omega(t+\tau)-\omega(t))^2 = 2(\rho(0)-\rho(\tau)) = \nu^2|\tau|^{2H}$: increments are those of an fBM [5], so paths are α-Hölder for every $\alpha < H$ (Kolmogorov–Chentsov). H thus controls roughness, T memory, and λ^2 intermittency.

Normalisation. Choosing $m = -\frac12\rho(0) = -\frac{\nu^2}{4}T^{2H}$ gives, by the Gaussian moment generating function, $\mathbb{E}e^{\omega(t)} = e^{m+\rho(0)/2} = 1$.

3. The log S-fBM measure and its MRM limit

Definition 2. *The log S-fBM random measure is $M_{H,T}(dt) = \sigma^2 e^{\omega_{H,T}(t)}dt$; for $I \subset \mathbb{R}$, $M_{H,T}(I) = \sigma^2\int_I e^{\omega_{H,T}(t)}dt$.*

By Fubini and the normalisation, $\mathbb{E}M_{H,T}(I) = \sigma^2|I|$: σ^2 is the mean variance per unit time, and $M_{H,T}$ is the model of integrated variance $\int_I \sigma_s^2 ds$.

Limit $H \to 0$. With λ fixed, $\nu^2 \to \infty$, $m \to -\infty$, but for $0 < \tau < T$

$$\rho(\tau) = \lambda^2\,\frac{T^{2H} - \tau^{2H}}{2H(1-2H)} \xrightarrow[H\to 0]{} \lambda^2 \ln\frac{T}{\tau}, \tag{2}$$

since $(T^{2H} - \tau^{2H})/2H \to \ln(T/\tau)$. This is exactly the logarithmic covariance of the log-normal MRM. Wu, Muzy and Bacry [8] show that $M_{H,T} \Rightarrow M_{0,T}$ weakly, the MRM being defined as a limit of exponentials of regularised log-correlated fields [4, 1]. The limit is non-degenerate iff $\lambda^2 < 2$, and for q with $\zeta(q) > 1$,

$$\mathbb{E}\big[M_{0,T}([0,\tau])^q\big] = K_q\,\tau^{\zeta(q)}, \quad \zeta(q) = q - \tfrac{\lambda^2}{2}q(q-1), \tag{3}$$

$\tau \le T$: the scaling is *exactly* multifractal (concave ζ), whereas for $H > 0$ it is only approximately so on scales $\tau \gg Te^{-1/(2H)}$. H therefore acts as a regulariser that breaks exact scale invariance while keeping stationarity.

4. Second-order moments

Let $M_\Delta(t) = M_{H,T}([t, t+\Delta])$ be the integrated variance over a sampling period Δ, and $C_M(\Delta,\tau) = \mathbb{E}[M_\Delta(t)M_\Delta(t+\tau)]$.

Proposition 1. *Let F solve $F''(z) = \sigma^4 e^{\rho(z)}$, $F(0) = F'(0) = 0$. Then*

$$C_M(\Delta,\tau) = F(\tau+\Delta) + F(|\tau-\Delta|) - 2F(\tau). \tag{4}$$

Proof. Since $\mathbb{E}e^{\omega(u)+\omega(v)} = e^{2m+\rho(0)+\rho(u-v)} = e^{\rho(u-v)}$, $C_M = \sigma^4\int_0^\Delta\int_0^\Delta e^{\rho(\tau+u-v)}du\,dv = \int_{-\Delta}^{\Delta}(\Delta - |s|)F''(\tau+s)\,ds$, and two integrations by parts give (4) (F'' is even, F is extended as an even function). □

For $H = 0$ and $z \le T$, $e^{\rho(z)} = (T/z)^{\lambda^2}$, so $F(z) = \sigma^4 T^{\lambda^2} z^{2-\lambda^2}/\big((2-\lambda^2)(1-\lambda^2)\big)$; for $H > 0$, F is expressed with lower incomplete gamma functions [8]. For $\tau \ge T + \Delta$, independence gives $C_M = \sigma^4\Delta^2$.

Log-variance. For small intermittency, $\ln M_\Delta(t) \approx \ln(\sigma^2\Delta) + m + \frac{1}{\Delta}\int_t^{t+\Delta}\omega$, hence, with $G'' = \rho$, $G(0) = G'(0) = 0$, the same computation yields

$$C_{\ln M}(\Delta,\tau) \approx \frac{G(\tau+\Delta) + G(|\tau-\Delta|) - 2G(\tau)}{\Delta^2}, \tag{5}$$

where, for $z \le T$, $G(z) = \frac{\nu^2}{2}\big(\frac{T^{2H}z^2}{2} - \frac{z^{2H+2}}{(2H+1)(2H+2)}\big)$. For $\Delta \to 0$ this reduces to $\rho(\tau)$: the RFSV structure is recovered at high frequency.

5. Estimation

Bias of scaling estimators. The usual estimator regresses $\ln m(q,\tau)$, $m(q,\tau) = \mathbb{E}|\ln M_\Delta(t+\tau) - \ln M_\Delta(t)|^q$, on $\ln\tau$. Under log S-fBM, $\ln m(q,\tau) \approx qH\ln\tau + \frac{q}{2}\ln g_H(\Delta/\tau)$, where g_H accounts for the averaging over

$[t, t+\Delta]$. The second term is not constant in τ, so the slope is biased upward; in our simulations a true $H = 0.002$ was estimated at 0.082. Data that are in fact multifractal can therefore look "rough".

GMM. Let $\theta = (H, \lambda^2, T, \sigma^2)$ and observe $M_\Delta(j\Delta)$, $j = 1, \dots, N$, $L = N\Delta$. Stack empirical moments $\widehat{C}_M(\Delta, k\Delta) = \frac{1}{N-k}\sum_j M_\Delta(j\Delta)M_\Delta((j+k)\Delta)$ and $\widehat{C}_{\ln M}(\Delta, k\Delta)$ for $k \in \mathcal{K}$ into $\hat{\boldsymbol{c}}$, and set

$$\hat{\theta} = \arg\min_{\theta} \big(\hat{\boldsymbol{c}} - \boldsymbol{c}(\theta)\big)^\top W \big(\hat{\boldsymbol{c}} - \boldsymbol{c}(\theta)\big). \tag{6}$$

Under ergodicity and identifiability, $\hat{\theta}$ is consistent and asymptotically normal, efficient for W the inverse long-run covariance of the moments [3]. Two regimes matter: in the *low-frequency* regime $\Delta < T \ll L$, all four parameters are identifiable; in the *high-frequency* regime $\Delta \ll L = O(T)$, typical of daily data over a few years, T and σ^2 are entangled (only σ^2 times a function of T is seen), but H and λ^2 remain identifiable, so no unrealistic ergodicity in T is needed.

Simulation. Exact simulation on n points by Cholesky costs $O(n^3)$; since ρ has compact support, circulant embedding [7] costs $O(n \log n)$ by FFT, provided the eigenvalues of the embedding circulant are non-negative, which we check numerically. Monte-Carlo moments agree with (4), and the convergence to MRM scaling is visible for $H \leq 0.05$.

6. Status

Implemented: simulation, the MRM limit, C_M and $C_{\ln M}$ validated against Monte-Carlo, bias of the scaling estimator. Next: full GMM calibration on daily realised volatility of individual stocks and indices, to decide whether data are best described by $H > 0$ (RFSV) or $H = 0$ (MRM).

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